Morphological alignment of tokenizers: working notes

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1 Where this started

Two observations from the boundary paper:

1. **Probing a word on its own is not neutral.** MorphScore and MorphAlign both tokenize the gold word without context. Under a leading-space vocabulary the bare string `leads` is not what the tokenizer sees in running text (`leads`), so it is split (`lead|s`) where in context it is one token. English plain BPE: 7.9% of MorphScore words are a single token on the bare probe, 64.7% in context.
2. **How whole-word tokens are credited decides the sign of the comparison.** MorphScore either drops single-token words (`exclude_single_tok=True`) or gives them a full point. The credited variant tracks single-token share at r = 0.95 across the grid.

Both metrics are weak in other ways, and existing gold is sparse and inconsistent, which led to the question in Section 3: can we build our own silver gold for the most frequent words of a corpus?

2 Metrics in use

2.1 MorphScore

Arnett & Bergen (COLING 2025) and Arnett, Hudspeth & O'Connor (TokShop 2025, 70 languages). Gold comes from Universal Dependencies: the stem is the substring of the word form shared with the lemma, the remainder is a prefix and/or suffix, so each item has one or two boundaries and is mostly inflectional. Scoring is per word, macro-averaged; recall is 1 if all gold boundaries are hit, 0.5 if one of two.

Bug, upstream and in our port (fixed in `catherinearnett/morphscore#3`, and in `script_bpe/analysis/morphscore.py` on this branch): the end of the word was counted as a predicted boundary, so a perfect `lead|s` scored precision 1/2 and an unsplit word (credited) scored 1. Recall is unaffected. Re-scoring the English boundary grid moves precision and F1 up, mostly for `plain`, and changes no ranking.

2.2 MorphAlign

Stephen & Libovický (arXiv 2601.18536), the `eval/morph-tok-eval` submodule. Gold-free with respect to segmentation: an IBM Model 1 alignment between subwords and UD morpho-syntactic features, scored by how well subwords predict features. No notion of boundary position; an unsplit word can only score zero; the scale differs by about 30x between languages. Our MinGram-paper usage also probes the bare word.

2.3 Boundary P/R/F1 (Creutz & Lindén 2004)

Used by Uzan et al. (ACL 2024) and Gow-Smith et al. (EMNLP 2022). Micro precision and recall over word-internal boundaries. An unsplit word loses recall on its gold boundaries and adds nothing to precision, which

is the behaviour we want without a credit switch. Gow-Smith argue precision is the more meaningful number because storing frequent words whole is legitimate.

3 Silver gold for frequent words

3.1 Goal

A gold file of surface segmentations (morphs concatenate to the word) for the top 10k-100k words of a web corpus, per language, with boundary types so a metric can choose its depth.

3.2 Background: Morfessor

Morfessor (Creutz & Lagus 2002, 2007; Morfessor 2.0, Virpioja et al. 2013) is the standard unsupervised morphological segmenter. It models a corpus as a sequence of morphs drawn from a lexicon and picks the lexicon by a two-part cost, in MDL or MAP form:

$$L(\theta, D) = \underbrace{-\log P(\theta)}_{\text{lexicon cost}} + \underbrace{-\log P(D \mid \theta)}_{\text{corpus cost}}$$

The corpus cost is a unigram model over morphs, exactly as in the Unigram LM tokenizer. The lexicon cost charges for every character of every morph type in the lexicon, so adding a morph must pay for itself by shortening the corpus encoding. **Morfessor Baseline** optimizes this greedily: for each word type it tries every binary split recursively and keeps the cheapest; segmentation of new words is Viterbi over the learned lexicon.

Relation to our tokenizers: Unigram LM (SentencePiece) has the same corpus model but replaces the lexicon prior with a hard vocabulary size and EM plus pruning. Morfessor has no target vocabulary size; the trade-off is set by a corpus weight α multiplying the corpus cost. Trained on token frequencies it keeps frequent words whole (they are cheap to store); trained on type counts, or log-dampened counts as here, it segments more.

Variants: **semi-supervised Morfessor** adds a likelihood term for annotated segmentations with its own weight; **FlatCat** (Grönroos et al. 2014) adds an HMM over morph categories (prefix, stem, suffix, non-morph) and is among the best on Morpho Challenge in semi-supervised mode.

Caveat raised during this work: silver gold learned with Morfessor from the same kind of corpus would reward tokenizers for behaving like Morfessor, which favours Unigram and MinGram. Morfessor is therefore only a baseline and a CRF feature here, never the source of gold.

3.3 Existing gold and how it is built

- **Morpho Challenge 2005-2010**: about 1000 randomly sampled words per language (en, fi, de, tr), canonical analyses from CELEX and FINTWOL.
- **SIGMORPHON 2022**: canonical segmentation, Wiktionary-derived, 9 languages (no de, fi). Winner 97.3 macro F1 (char seq2seq); Morfessor baseline 25.6 on this canonical task.
- **UniSegments 1.0**: 38 harmonised datasets, 30 languages, licences per dataset.
- **MorphyNet**: Wiktionary, 15 languages, mixed surface and lemma-based segmentations.
- **CRF boundary taggers** (Ruokolainen et al. 2013, 2014): 1000 annotated words give 85.9 (en) and 83.8 (fi) boundary F1; adding Morfessor features gives 88.4 and 89.4.
- **LLM annotation**: thin evidence. Prompted frontier LLMs are "inconsistent" (Morozov et al. 2026); Claude 3.7 under-segments Bribri. No large silver set exists.

3.4 Data

Word frequencies: first 1 GB of each Goldfish corpus, lowercased letter runs (`word_counts.py`). English: 168M tokens, 1.1M types.

Reference: SIGMORPHON 2022 English word gold, projected from canonical to surface by character alignment (`gold.py`), surface-only letters attached to the left morph. 70% of English rows concatenate exactly; Russian only 11% (lemma-based), so the projection is shaky there.

Coverage of the most frequent words is poor:

top-k	SIGMORPHON	CELEX (morph-tok-eval)
1k	26%	70%
10k	71%	51%
100k	63%	19%

and the sources disagree: CELEX keeps `leads` and `understanding` whole, SIGMORPHON gives `lead|s` and `under|stand|ing`; SIGMORPHON has no `running` or `walked`.

3.5 Experiments

3.5.1 Classical segmenters against the SIGMORPHON reference

English, Goldfish frequency bands. Cells are boundary F1 (precision/recall) against SIGMORPHON 2022 projected to surface. Supervised methods train on SIGMORPHON words outside the top 50k. `CELEX` is not a method: it is the second human-derived gold, scored against the first, so its row is the agreement between two references.

	top 1k	1k-10k	10k-50k
CELEX vs SIGMORPHON	0.54 (0.86/0.39)	0.67 (0.72/0.62)	0.74 (0.75/0.72)
no split	0.00 (1.00/0.00)	0.00 (1.00/0.00)	0.00 (1.00/0.00)
Morfessor, $\alpha = 1$	0.12 (1.00/0.06)	0.22 (0.96/0.12)	0.31 (0.95/0.18)
Morfessor, $\alpha = 0.3$	0.54 (0.81/0.40)	0.66 (0.80/0.57)	0.64 (0.64/0.65)
Morfessor, $\alpha = 0.1$	0.58 (0.51/0.66)	0.60 (0.50/0.75)	0.53 (0.41/0.76)
Morfessor, $\alpha = 0.03$	0.35 (0.23/0.76)	0.41 (0.28/0.78)	0.41 (0.28/0.79)
Morfessor semi-sup. (1k)	0.59 (0.73/0.50)	0.59 (0.77/0.48)	0.62 (0.79/0.51)
CRF (1,000)	0.67 (0.69/0.65)	0.71 (0.75/0.67)	0.70 (0.78/0.63)
CRF + Morfessor (1,000)	0.66 (0.72/0.61)	0.71 (0.82/0.62)	0.66 (0.83/0.55)
CRF (10,000)	0.73 (0.71/0.77)	0.78 (0.78/0.78)	0.78 (0.81/0.75)
CRF + Morfessor (10,000)	0.73 (0.73/0.72)	0.78 (0.82/0.75)	0.77 (0.84/0.71)
CRF (50,000)	0.78 (0.77/0.80)	0.85 (0.84/0.85)	0.84 (0.86/0.83)
CRF + Morfessor (50,000)	0.79 (0.78/0.80)	0.85 (0.86/0.83)	0.83 (0.87/0.80)

Unsupervised Morfessor tops out at 0.67 F1 at its best corpus weight, which is the range the literature reports. A CRF with Ruokolainen features learns from labelled words steadily (0.71 at 1k, 0.85 at 50k); Morfessor features buy precision for recall and no F1. The two human-derived references agree with each other at only 0.67 F1 in the 1k-10k band, so 0.85 against one of them is close to what any method can show on this yardstick.

3.5.2 LLM annotation

Prompt: typed surface segmentation (`llm_segment.TYPED_PROMPT`), boundaries marked + inflection, - derivation, = compound, with explicit conventions for depth (split all the way down), bound roots (split an affix only if the base is a real related word) and spelling changes (affixes keep their dictionary form, the stem absorbs the change: `ignor+ed`, `patch+es`). An earlier untyped prompt left depth open and Sonnet split only the outer layer (F1 0.78 against SIGMORPHON, against 0.85 typed).

Pairwise boundary F1 on 290 words drawn from the top 10k and present in SIGMORPHON. F1 is symmetric, so each cell reads as agreement. Claude models run through `claude -p`, the others through the Copilot CLI.

	Haiku	Haiku (run 2)	Sonnet	Gemini 3.8 Flash	GPT-6 Astra	SIGMORPHON	CRF (50k)	CELEX
Haiku	–	0.95	0.93	0.95	0.93	0.85	0.83	0.71

Haiku (run 2)	0.95	–	0.93	0.96	0.93	0.86	0.84	0.74
Sonnet	0.93	0.93	–	0.97	0.92	0.85	0.82	0.70
Gemini 3.8 Flash	0.95	0.96	0.97	–	0.94	0.86	0.83	0.73
GPT-6 Astra	0.93	0.93	0.92	0.94	–	0.86	0.81	0.74
SIGMORPHON	0.85	0.86	0.85	0.86	0.86	–	0.86	0.72
CRF (50k)	0.83	0.84	0.82	0.83	0.81	0.86	–	0.68
CELEX	0.71	0.74	0.70	0.73	0.74	0.72	0.68	–

LLMs agree with each other at 0.93-0.97, across model families as much as within one, and with either human-derived reference at 0.70-0.86. Two runs of the same model agree at 0.95-0.97, so sampling noise is a small part of the disagreement. By boundary type, inflection is almost settled and derivation is where the remaining disagreement lives:

boundary type	mean pairwise F1 (range)
inflection	0.98 (0.96-1.00)
derivation	0.88 (0.83-0.94)
compound	0.92 (0.90-1.00)

Of the 256 words where all four models (Haiku, Sonnet, Gemini, GPT; three families) agree exactly, 29 differ from SIGMORPHON. Most are reference errors (`alex|a`, `gen|ome`, `line|n`, `los|t`, `give|a|way`) or artefacts of projecting canonical gold to surface (`ignore|d`, `patche|s`, `consume|r|s`, where the LLM convention is `ignor+ed`, `patch+es`, `consum-er+s`). A few go the other way and are debatable deep Latinate splits (`typ-ic-al`, `athlet-ic-s`). The 35 words where the LLMs disagree among themselves are bound roots and depth (`cognit-ive`, `re-mind+ed`, `proto-type`), which a per-boundary vote resolves.

3.5.2.1 Does the rulebook make the agreement?

The typed prompt fixes depth, bound roots and spelling conventions, so models that follow it will agree partly because of the rules. Rerun with a minimal prompt ("split each word into its morphemes, pieces must concatenate to the word", `llm_segment.MINIMAL_PROMPT`):

	Haiku	Sonnet	Gemini	GPT	SIGMORPHON	CELEX
Haiku	–	0.84	0.81	0.84	0.78	0.65
Sonnet	0.84	–	0.89	0.87	0.79	0.71
Gemini	0.81	0.89	–	0.93	0.85	0.79
GPT	0.84	0.87	0.93	–	0.82	0.75
SIGMORPHON	0.78	0.79	0.85	0.82	–	0.73
CELEX	0.65	0.71	0.79	0.75	0.73	–

Agreement drops from 0.93-0.97 to 0.82-0.93: the rulebook accounts for roughly 0.08 of it. The remaining minimal-prompt disagreements are the same convention questions the rulebook settles, not different morphology: etymological roots (Haiku's `con|tain`, `in|sert`, `e|merge`), e-drop (`ignore|d` against `ignor|ed`), consonant doubling (`fit|ting` against `fitt|ing`) and depth (`behavior|al` against `behav|ior|al`). Gemini is the strongest minimal-prompt annotator (0.85 against SIGMORPHON, the same as typed); Haiku is the weakest (0.78) because it splits Latinate roots.

Caveat: agreement is not accuracy. The models may share biases, the human references use different conventions from each other and from the rulebook, and the SIGMORPHON reference here is a projection that adds its own errors. A hand check of a random sample is the only direct accuracy measurement and has not been done yet.

4 Open questions

- Which boundary depth should a tokenizer metric use: inflection only, all, or report each?
- Paradigm consistency (does `lead+s` reuse `lead`?) as a metric that needs no segmentation.
- In-context probing for every metric.